M1 Project Report

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Course: AI and ML for Industry 05

Contents

Abstract

The core tasks involve **eigen decomposition**, applying **Principal Component Analysis (PCA)** for dimensionality reduction, and assessing the reconstruction error using **Peak Signal-to-Noise Ratio (PSNR)**.

Analysed a subset of the test dataset and successfully demonstrated how PCA effectively reduces dimensionality while retaining critical information.

Notebook

Notebook is attached.

module1Project.ipynb

Tasks

The project is structured into four main tasks to demonstrate the concepts:

- **Task 1: Eigen Decomposition** Compute the eigen decomposition of the sample covariance matrix. The eigenvalues are used to calculate the percentage of variance explained. A plot of the cumulative sum of these percentages versus the number of components is generated.
- **Task 2: PCA for Dimensionality Reduction** Apply PCA via eigen decomposition to reduce the dimensionality of the images for $p \in \{50,250,500\}$.
- **Task 3: Data Reconstruction** Using the reduced data from Task 2, reconstruct the original images. This is done by leveraging the property of orthonormal matrices.
- **Task 4: Error Comparison (PSNR)** Compare the error between the original and reconstructed images for 5 randomly selected images using the **Peak Signal-to-Noise Ratio (PSNR)**.

Task1: Eigen Decomposition

The plot of cumulative variance vs the number of components of the values calculated is shown on next page.

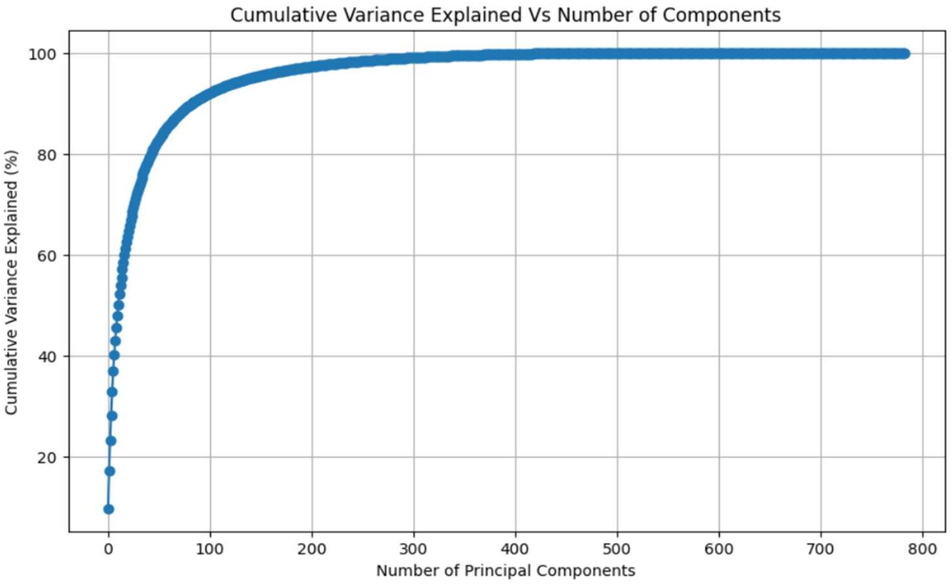

Fig 1. Cumulative Variance explained vs number of components.

The first 50 components explain a certain percentage of the variance almost 80%, and this percentage increases substantially as we include more components, such as 250 or 500. This confirms that the most important features of the images can be represented in a lower-dimensional space.

Around 250 components 100% of cumulative variance is achieved.

Task2: PCA for Dimensionality Reduction

PCA is applied via eigen decomposition to reduce the dimensionality of images.

```
Reduced data shape with 50 components: (2000, 50)
Reduced data shape with 250 components: (2000, 250)
Reduced data shape with 500 components: (2000, 500)
```

Fig2. PCA Applied

We see that the components get reduced by the value of chosen p.

Task3: Data Reconstruction

Using the reduced data from above task, image is reconstructed and the results are as shown on next page.

Original image is chosen based on index 100. We can choose another index too.

Original Image Index 100

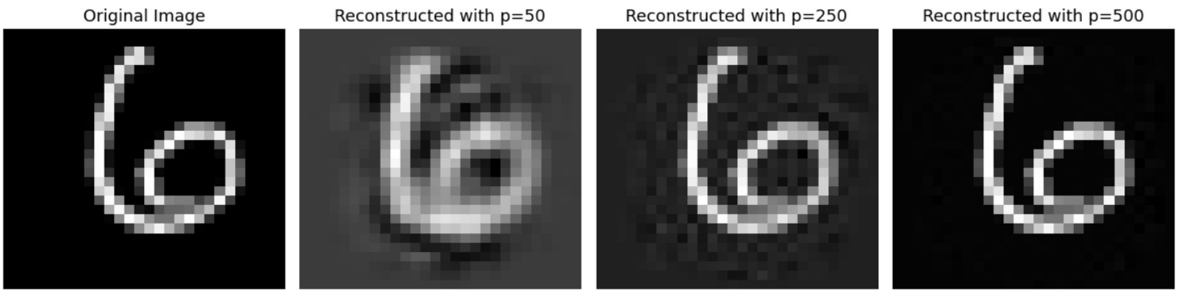

Fig3. Original Image Index 100 and the reconstructed ones with different p values.

For Original image index 10

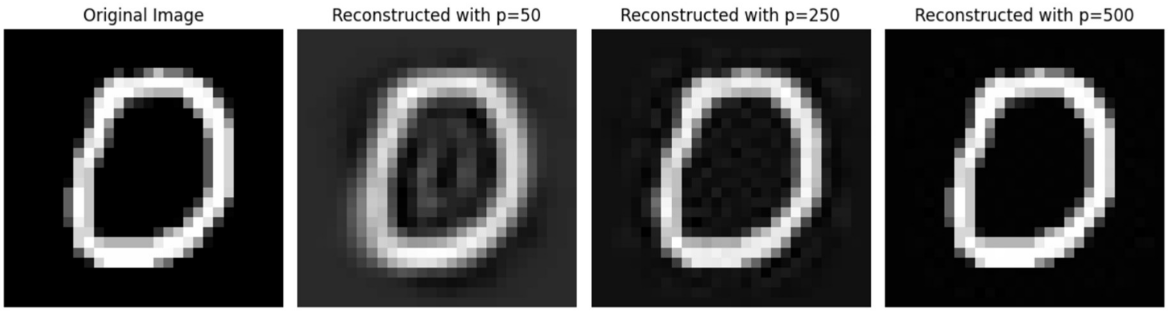

Fig 4. Original Image Index 10 and reconstructed ones with different p values.

Task4: Error Comparison using PSNR

In this task we compare the error between the original and reconstructed images for 5 randomly selected images using Peak Signal-to-Noise Ratio (PSNR).

Function used for calculation of PSNR is taken from the reference mentioned in assignment and is as given below.

```
def PSNR(original, compressed):
    mse = np.mean((original - compressed) ** 2)
    if(mse == 0):  # MSE is zero means no noise is present in the signal .
                  # Therefore PSNR have no importance.
        return 100
    max_pixel = 255.0
    psnr = 20 * log10(max_pixel / sqrt(mse))
    return psnr
```

Fig 5. PSNR Function

The PSNR values are calculated for each value of p i.e. {50, 250, 500} for randomly selected images based on indices. Indices is generated randomly.

Below is the randomly selected image indices for this run.

```
Randomly selected image indices for PSNR comparison: [1292 1350 1856 1723 1509]
```

Fig 6. Randomly selected image indices

Every time we run this cell of notebook, we would get different randomly selected indices as base seed has not been selected.

PSNR values for p 50 are as given below:

```
PSNR values for reconstructed images with p=50:
Image index 1292: PSNR = 69.14 dB
Image index 1350: PSNR = 73.26 dB
Image index 1856: PSNR = 65.26 dB
Image index 1723: PSNR = 69.61 dB
Image index 1509: PSNR = 66.27 dB
```

Fig 7. PSNR values for p=50

PSNR values for p 250 are as given below:

```
PSNR values for reconstructed images with p=250:
Image index 1292: PSNR = 78.55 dB
Image index 1350: PSNR = 86.23 dB
Image index 1856: PSNR = 75.72 dB
Image index 1723: PSNR = 78.65 dB
Image index 1509: PSNR = 76.21 dB
```

Fig 8. PSNR values with p=250

And the PSNR values for p 500 are as given below:

```
PSNR values for reconstructed images with p=500:
Image index 1292: PSNR = 101.25 dB
Image index 1350: PSNR = 109.03 dB
Image index 1856: PSNR = 92.90 dB
Image index 1723: PSNR = 101.29 dB
Image index 1509: PSNR = 97.69 dB
```

Fig 9. PSNR values with p=500

Summary of PSNR values for different p values is as given below:

```
Summary of PSNR values for different p values:
p=50: PSNR values = [69.13520065865112, 73.26384208866995, 65.25997134026485, 69.60894906357355, 66.2713558016691]
p=250: PSNR values = [78.54871920064866, 86.23378876891393, 75.71731774289043, 78.6466440650833, 76.21357090688822]
p=500: PSNR values = [101.25491145250847, 109.02650528458327, 92.90231178824527, 101.29049832314553, 97.68858729884803]
```

Fig 10. Summary of PSNR values

Results and Discussion

R1. Variance Explained

The cumulative variance explained by the principal components shows that a large amount of the variance can be captured with a smaller number of components.

R2. Data Reconstruction

The reconstructed images for different values of p show varying levels of detail, with higher values of p leading to better reconstruction.

As p increases, the reconstructed images become sharper and more closely resemble the originals.

R3. Error Comparison

The PSNR values, which indicate reconstruction quality, show that higher PSNR values correspond to less reconstruction error.

The PSNR values increase with higher values of p, confirming that more principal components lead to better image reconstruction.

Conclusion

In this project, we successfully applied PCA to the MNIST dataset, effectively reducing its dimensionality while retaining most of the variance. The reconstructed images showed good quality with minimal error, as supported by the PSNR values. This demonstrates the effectiveness of PCA as a tool for dimensionality reduction and data representation in machine learning.